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Supplementary Materials for

Fragment binding to the Nsp3 macrodomain of SARS-CoV-2 identified through crystallographic screening and computational docking

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The PDF file includes:

Supplementary Text
Figs. S1 to S9
Legends for data files S1 and S2
References

Other Supplementary Material for this manuscript includes the following:

(available at advances.sciencemag.org/cgi/content/full/7/16/eabf8711/DC1)

Data files S1 and S2

Supplementary Text

Purity and structure determination of fragments ZINC901381520, ZINC82473428 and ZINC89254160 from Enamine

Samples of ZINC901391520, ZINC82473428 and ZINC89254160 obtained from Enamine were expected to be N_9-alkylated isomers but electron density of the fragments in X-ray structures indicated these samples were N_3-alkylated isomers instead (ZINC901391520_N3, ZINC82473428_N3 and ZINC89254160_N3, see **Fig. S7**I-L). The original samples of ZINC901391520, ZINC82473428 and ZINC89254160 used in fragment screening by X-ray crystallography were analyzed by HPLC-MS and ^{1}H NMR to confirm sample purity and corroborate structure. There is no reported characterization data to be used as reference for structure confirmation for the N_9- or N_3-alkylated compounds ZINC901391520 and ZINC89254160. The N_9-alkylated structure ZINC82473428 has previously been prepared and tabulated NMR data has been reported (*82*).

A re-supplied sample of ZINC901391520 from a new batch synthesized at Enamine was confirmed by ^{1}H NMR to be >95% purity and a different isomer than the original sample of ZINC901391520. The X-ray crystal structure of this fragment in complex with Mac1 revealed the fragment to be the N_9-alkylated isomer (**Fig. S7**I).

The original samples of ZINC901391520, ZINC82473428 and ZINC89254160 from Enamine used in the fragment screen were evaluated for purity by HPLC on an Agilent 1200 Binary SL system with diode array detection and mass spectrometric detection on an Agilent 6135B Quadrupole system in electrospray ionization mode (positive ion detection). One of two HPLC Methods A or B were used to determine sample purity using mobile phase linear gradients of acetonitrile with 0.1% TFA in water with 0.1% TFA. The flow rate was 1.000 ml/min through a Phenomenex Gemini 3 mm C18 110 Å LC column (4.6 mm dia. x 150 mm length).

HPLC Method A mobile phase gradient: Gradient time points (minutes): 1.0-1.5-10.5-11.0-12.5-13.0-15.0; % acetonitrile at gradient time points: 5-5-20-95-95-5-5

HPLC Method B mobile phase gradient: Gradient time points (minutes): 1.0-7.0-8.0-10.0-10.5-12.0; % acetonitrile at gradient time points: 5-30-95-95-5-5

NMR experiments for samples ZINC901391520, ZINC82473428 and ZINC89254160

Original samples of ZINC901391520, ZINC82473428 and ZINC89254160 from Enamine used in the fragment screen were dissolved in d_6-DMSO and analyzed by ^{1}H and ^{13}C NMR on a Bruker 400 MHz instrument with Avance III electronics. Data was obtained at ambient temperature (ca. 25°C) collecting 64 scans for proton experiments and 1024 scans for carbon experiments. Raw data was processed and reports created using ACD Spectrus software.

Original sample ZINC901391520

A sample of 5.5 mg ZINC901391520 was dissolved in 0.75 ml d_6-DMSO for NMR analysis and from this solution 50 µl was diluted in 0.45 ml acetonitrile to make up the analytical sample for HPLC-MS using HPLC Method A. The sample chromatogram from HPLC revealed a single peak with UV absorbance at both 214 and 254 nm at t_R = 5.272 minutes. Aside from a very strong UV_{214} peak at t_R = 2.00 minutes attributed to DMSO cosolvent in the sample, no other

peaks were observed at these UV wavelengths and the sample purity was estimated to be >98% based on UV peak area. ^{1}H NMR (400 MHz, *d*$_6$-DMSO, 25°C) δ ppm 8.49 (s, 1H), 7.91-8.26 (br d, 2H), 7.76 (s, 1H), 6.30 (s, 1H), 5.63 (s, 2H), 3.86 (s, 3H). ^{13}C NMR (101 MHz, *d*$_6$-DMSO, 25°C) δ ppm 172.05, 167.57, 155.01, 152.46, 149.39, 143.55, 120.18, 94.85, 57.13, 44.32. LRMS (ESI$^+$) for peak at t_R = 5.272 minutes: observed *m/z* = 247.3 $[MH]^+$ for $C_{10}H_{10}N_6O_2$ exact mass = 246.09.

Second batch sample ZINC901391520
A sample was dissolved in 0.75 ml *d*$_6$-DMSO for NMR analysis. ^{1}H NMR (400 MHz, *d*$_6$-DMSO, 25°C) δ ppm 8.24 (s, 1H), 8.16 (s, 1H), 7.31 (br s, 2H), 6.22 (s, 1H), 5.49 (s, 2H), 3.86 (s, 3H).

Sample ZINC82473428
A sample of 3.9 mg ZINC82473428 was dissolved in 0.75 ml *d*$_6$-DMSO for NMR analysis and from this solution 50 µl was diluted in 0.45 ml acetonitrile to make up the analytical sample for HPLC-MS using HPLC Method B. The sample chromatogram from HPLC revealed a single peak with UV absorbance at both 214 and 254 nm at t_R = 3.766 minutes. Aside from a very strong UV_{214} peak at t_R = 2.00 minutes attributed to DMSO cosolvent in the sample, no other peaks were observed at these UV wavelengths and sample purity was estimated to be >98% based on UV peak area. ^{1}H NMR (400 MHz, *d*$_6$-DMSO, 25°C) δ ppm 8.31 (s, 1H), 8.01 (br s, 2H), 7.86 (s, 1H), 4.44 (dd, *J*=13.18, 3.39 Hz, 1H), 4.31-4.40 (m, 1H), 4.20-4.30 (m, 1H), 3.75-3.87 (m, 1H), 3.58-3.70 (m, 1H), 1.93-2.07 (m, 1H), 1.75-1.92 (m, 2H), 1.58-1.73 (m, 1H). ^{13}C NMR (101 MHz, *d*$_6$-DMSO, 25°C) d ppm 154.78, 151.53, 149.56, 144.30, 75.35, 67.24, 52.54, 40.44, 28.23, 25.03. LRMS (ESI$^+$) for peak at t_R = 3.766 minutes: observed *m/z* = 220.3 $[MH]^+$ for $C_{10}H_{13}N_5O$ exact mass = 219.11.

Reported NMR data for compound ZINC82473428_N9 from Rad et al. (*82*): ^{1}H NMR (400 MHz, *d*$_6$-DMSO, 25°C) δ ppm 7.91 (s, 1H), 7.83 (s, 1H), 7.01 (br s, 2H), 3.87-3.99 (m, 3H), 3.34-3.52 (m, 2H), 1.30-1.54 (complex m, 4H). ^{13}C NMR (101 MHz, *d*$_6$-DMSO, 25°C) δ ppm 156.6, 152.9, 149.2, 144.7, 117.2, 80.6, 67.9, 57.8, 29.1, 25.1.

Sample ZINC89254160
A sample of 3.2 mg ZINC89254160 was dissolved in 0.75 ml *d*$_6$-DMSO for NMR analysis and from this solution 50 µl was diluted in 0.45 ml acetonitrile to make up the analytical sample for HPLC-MS using HPLC Method A. The sample chromatogram from HPLC revealed a major peak and a minor peak with UV absorbances at both 214 and 254 nm: major peak t_R = 6.530 minutes and minor peak t_R = 6.751 minutes. Relative peak area calculated as percentage of combined UV peak area at 254 nm was 93.3% major peak and 6.7% minor peak (corresponds to ca. 14:1 ratio). Aside from a very strong UV_{214} peak at t_R = 2.00 minutes attributed to DMSO cosolvent in the sample, no other peaks were observed at these UV wavelengths. Tabulated NMR data reported here for major peaks only. ^{1}H NMR (400 MHz, *d*$_6$-DMSO, 25°C) δ ppm 8.47 (s, 1H), 7.95 (br s, 2H), 7.73 (s, 1H), 7.47 (s, 1H), 5.55 (s, 2H), 2.60 (s, 3H). ^{13}C NMR (101 MHz, *d*$_6$-DMSO, 25°C) δ ppm 166.21, 154.93, 152.47, 149.63, 149.51, 143.63, 120.43, 117.69, 48.08, 18.66. LRMS (ESI$^+$) for major peak at t_R = 6.530 minutes: observed *m/z* = 247.3 $[MH]^+$ for $C_{10}H_{10}N_6S$ exact mass = 246.07. LRMS (ESI$^+$) for minor peak at t_R = 6.751 minutes: observed *m/z* = 247.3 $[MH]^+$ for $C_{10}H_{10}N_6S$ exact mass = 246.07.

Major peak and minor peak have the same observed mass peak in LRMS and are presumed to be different N-alkylated isomers.

Conclusions based on HPLC-MS and NMR characterization of samples ZINC901391520, ZINC82473428 and ZINC89254160

HPLC-MS data confirmed that samples ZINC901391520 and ZINC82473428 are single compounds with >98% purity and mass peaks corresponding to either N_9- or N_3-alkylated isomers. Both ^{1}H and ^{13}C NMR data corroborated that the initial samples ZINC901391520 and ZINC82473428 were single compounds with a purity >98%. The very high purity determined for these two samples rules out the possibility that the structures determined by X-ray crystallography were the result of trace amounts of the alternative isomer in the samples. For ZINC89254160, HPLC-MS data confirmed that there was a 13:1 ratio of isomers in this sample and it is possible that the X-ray crystal structure of Mac1 obtained with ZINC89254160 was the result of protein complexed to trace/minor amounts of the alternative isomer (N3-alkylated).

The NMR data obtained for sample ZINC82473428 used in the crystallographic fragment screen does not match NMR data reported in the literature for the N_9-alkylated ZINC82473428 and thus this sample is presumed not to be the N_9-alkylated isomer. NMR data is not sufficient to unambiguously assign N_3- or N_9-alkylated structures for ZINC901391520, ZINC82473428 or ZINC89254160 and the unambiguous structure assignment of ZINC901391520, ZINC82473428 and ZINC89254160 as N_3-alkylated isomers in this work was provided by the electron density observed for these fragments in the Mac1 X-ray crystal structures.

The crystal structure of Mac1 with ZINC400552187 also revealed the N_3-alkylated structure instead of the requested N_9-alkylated form. Using DSF and ITC, ZINC901391520, ZINC82473428, ZINC89254160, ZINC400552187 were initially screened as the N_3-alkylated isomer (ZINC901391520_N3 (PDB 5RSK), ZINC82473428_N3 (PDB 5RVF), ZINC89254160_N3 (PDB 5RSJ), ZINC400552187_N3 (PDB 5RVG)). In addition, the N_9-alkylated ZINC901391520 (PDB 5S6W) was tested in DSF and the peptide-competition assay (HTRF) (data file S1).

QCRG Structural Biology Consortium authorship

This work was supported by the QCRG (Quantitative Biosciences Institute Coronavirus Research Group) Structural Biology Consortium. Listed below are the contributing members of the consortium listed by teams. Within each team the team leads are italicized (responsible for organization of each team, and for the experimental design utilized within each team), then the rest of team members are listed alphabetically.

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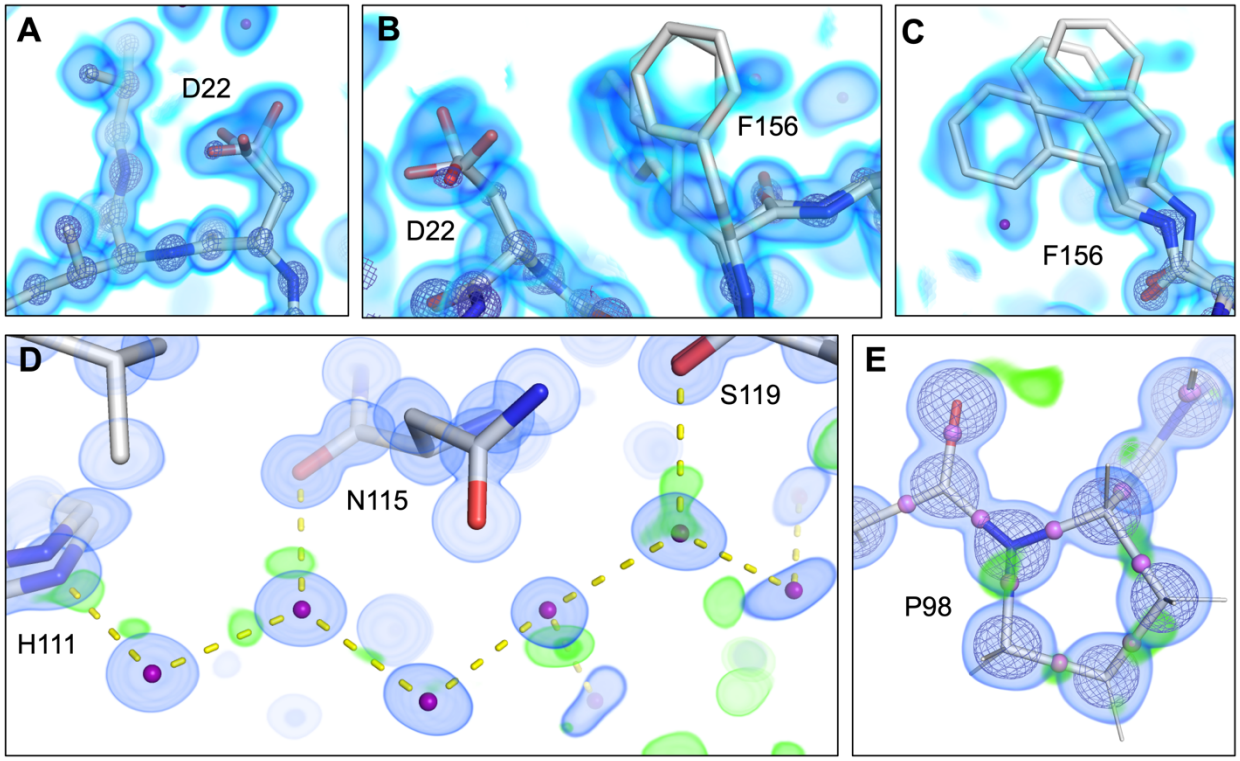
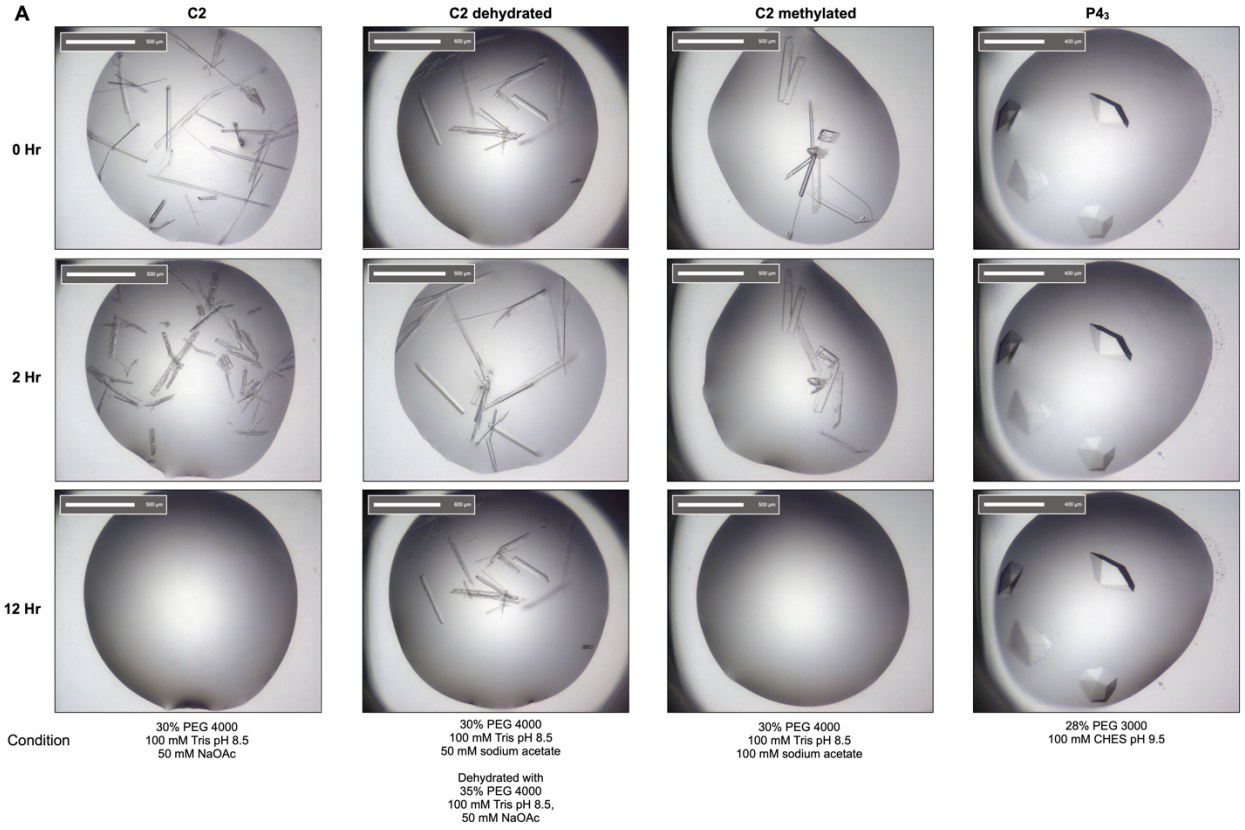
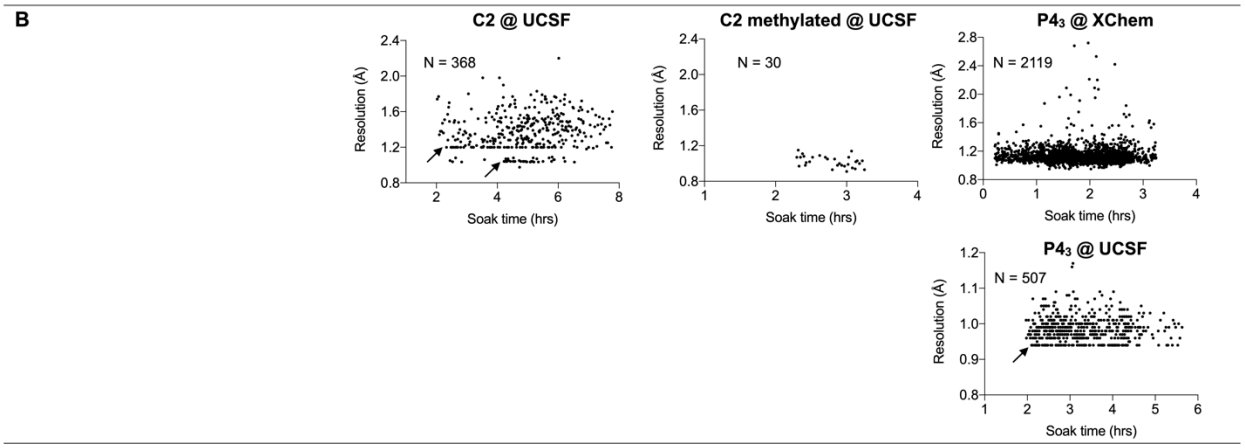


Figure S1. Ultra-high resolution features in Mac1 electron density maps.

A,B,C) Residues Asp22 and Phe156 both display conformational heterogeneity in high-resolution electron density maps of apo Mac1. The ultra-high resolution nature of the electron density is evident in (A), where atoms belonging to residues Ile23 and Leu24, as well as the backbone, appear as separate spheres at high contour. In (**A,B,C**) three $2mF_O$-DF_C maps are shown, contoured at 4.0 σ (blue mesh), 1.0 σ (blue volume), and 0.5 σ (cyan volume). **D)** Mac1 electron density reveals ordered water networks, including difference density that suggests the positions of hydrogen atoms within the water network. In (**D**), a $2mF_O$-Df_C map is shown, contoured at 1.5 σ (blue volume), and a mFo-DFc map is also shown, contoured at 2.5 σ (green volume). Hydrogen bonding interactions are depicted as dashed yellow lines. **E)** Two electron density maps are shown for P98, a $2mF_O$-DF_C map, contoured at 1.5 σ (blue volume), and a mF_O-DF_C map, contoured at 2.0 σ (green volume). The green peaks of positive difference density between heavy atom positions suggest covalent bonding densities, and we refined a model of Mac1 that included interatomic scatterers (IAS), shown in violet in the figure.


A
C2
C2 dehydrated
C2 methylated
P4₃
0 Hr
2 Hr
12 Hr
Condition
30% PEG 4000
100 mM Tris pH 8.5
50 mM NaOAc
30% PEG 4000
100 mM Tris pH 8.5
50 mM sodium acetate
Dehydrated with
35% PEG 4000
100 mM Tris pH 8.5,
50 mM NaOAc
30% PEG 4000
100 mM Tris pH 8.5
100 mM sodium acetate
28% PEG 3000
100 mM CHES pH 9.5


B
C2 @ UCSF
N = 368
C2 methylated @ UCSF
N = 30
P4₃ @ XChem
N = 2119
P4₃ @ UCSF
N = 507
Resolution (Å)
Soak time (hrs)

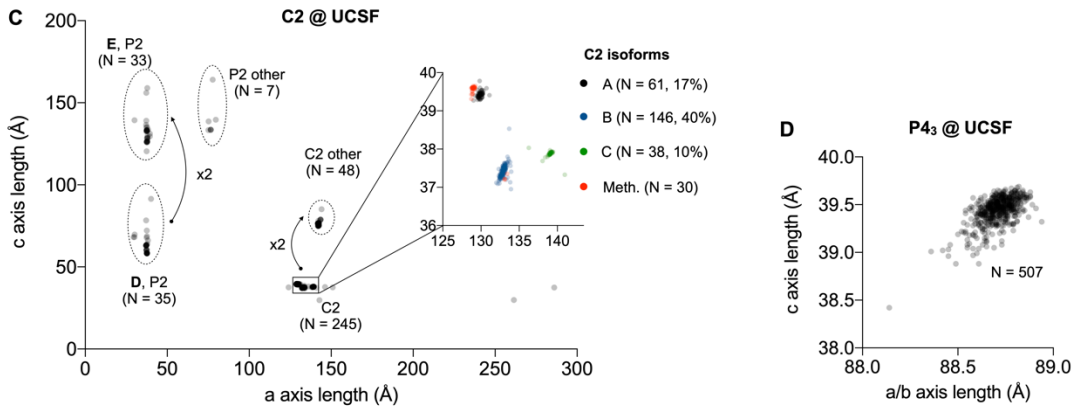

C
C2 @ UCSF
E, P2
(N = 33)
P2 other
(N = 7)
x2
D, P2
(N = 35)
C2 other
(N = 48)
x2
C2
(N = 245)
C2 isoforms
A (N = 61, 17%)
B (N = 146, 40%)
C (N = 38, 10%)
Meth. (N = 30)
c axis length (Å)
a axis length (Å)
D
P4₃ @ UCSF
N = 507
c axis length (Å)
a/b axis length (Å)

Figure S2. Comparison of isomorphism and DMSO tolerance of the C2 and $P4_3$ crystals. **A)** Images of crystals after soaking with 10% DMSO for 0, 2 and 12 hours. **B)** Resolution of the three crystal forms as a function of soak time for the datasets collected at XChem and UCSF. The arrows indicate where the measurement of high-resolution reflections was limited by the experimental setup. **C)** Multiple isoforms were observed for the C2 crystals after dehydration. Isoforms were distinguished based on differences in the a and c unit cell lengths. Arrows indicate where doubling of the a or c axis occurred. Inset: the majority of the datasets that were indexed in C2 (245, 84%) could be clustered into three isoforms (A, B and C). Of the 30 datasets collected for crystals grown from methylated protein, the majority (28) were similar to the A isoform. **D)** The $P4_3$ crystals were isomorphous.

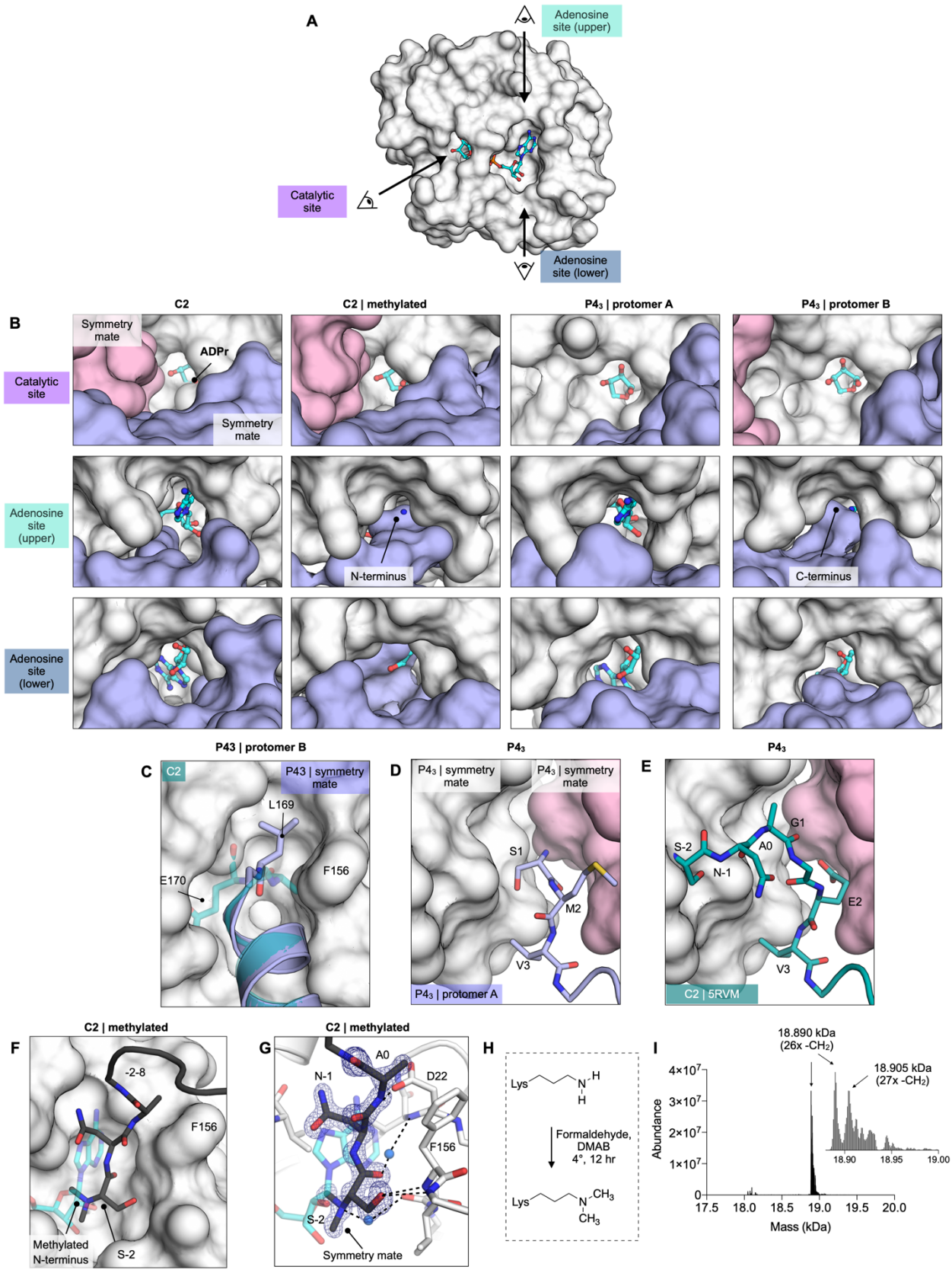

A
Adenosine site (upper)
Catalytic site
Adenosine site (lower)
B
C2
C2 | methylated
P4₃ | protomer A
P4₃ | protomer B
Symmetry mate
ADPr
Catalytic site
Adenosine site (upper)
Adenosine site (lower)
N-terminus
C-terminus
C
P43 | protomer B
C2
P43 | symmetry mate
L169
E170
F156
D
P4₃
P4₃ | symmetry mate
P4₃ | symmetry mate
S1
M2
V3
P4₃ | protomer A
E
P4₃
S-2
N-1
A0
G1
E2
V3
C2 | 5RVM
F
C2 | methylated
-2-8
F156
Methylated N-terminus
S-2
G
C2 | methylated
A0
N-1
D22
F156
S-2
Symmetry mate
H
Lys
Formaldehyde, DMAB 4°, 12 hr
Lys
I
18.890 kDa (26x -CH2)
18.905 kDa (27x -CH2)
Abundance
Mass (kDa)

Figure S3. Crystal packing in Mac1 crystals determines active site accessibility.
A) Active site access in the C2 and $P4_3$ crystals. Mac1 is shown as a white surface with ADP-ribose bound in the active site shown as cyan sticks. The three access points are indicated with arrows. **B)** Crystal packing defines the three access points. The catalytic site is partially obstructed in the C2 crystals, but open in both protomers of the $P4_3$ crystals. In both the methylated C2 crystals and protomer B of the $P4_3$ crystals, the adenosine site is obstructed. **C)** The C-terminal leucine (Leu169) of the $P4_3$ construct occupies the adenosine site of a symmetry mate. The adenosine site is shown as a white surface and the C-terminal residues are shown with blue sticks/cartoon. The C2 sequence (transparent teal cartoon/sticks) has an additional residue at the C-terminus (Glu170), and is therefore incompatible with the $P4_3$ crystal packing. **C)** The N-terminal residues of the $P4_3$ sequence (blue sticks) pack between two symmetry mates (white and pink surface). Compared to the $P4_3$ sequence, the C2 sequence contains a substitution (Met2Glu) and a three residue insertion (Asn-Ala-Gly). These residues were typically disordered, however, they were resolved in one of the fragment structures (ZINC157088 | 5RVM) (shown aligned to the $P4_3$ protomer A in **(E)**). Like differences in the C-termini, differences in the N-termini may have contributed to the distinct crystal packing seen for the two Mac1 structures reported in this work. **F)** The adenosine site was obstructed by a symmetry mate in the structure determined from crystals grown using methylated C2 protein. **G)** In the structure of methylated Mac1, the side-chain hydroxyl of Ser-2 occupies the oxyanion subsite. Electron density ($2mF_O$-DF_C) is shown as a blue mesh, contoured at 1.5 σ. **H)** Free amines were methylated using formaldehyde and dimethylamine borane (DMAB). The reaction is shown for lysine, however, based on the electron density shown in **(G)**. **I)** LC/MS analysis of methylated Mac1 (C2 construct). The mass spectrum was deconvoluted using MaxEnt1. The major peak (18.89 kDa) is consistent with the methylation of 13 lysine residues (26x -CH2). The minor peak (18.905 kDa + 15 Da) suggests that methylation was not 100% complete.

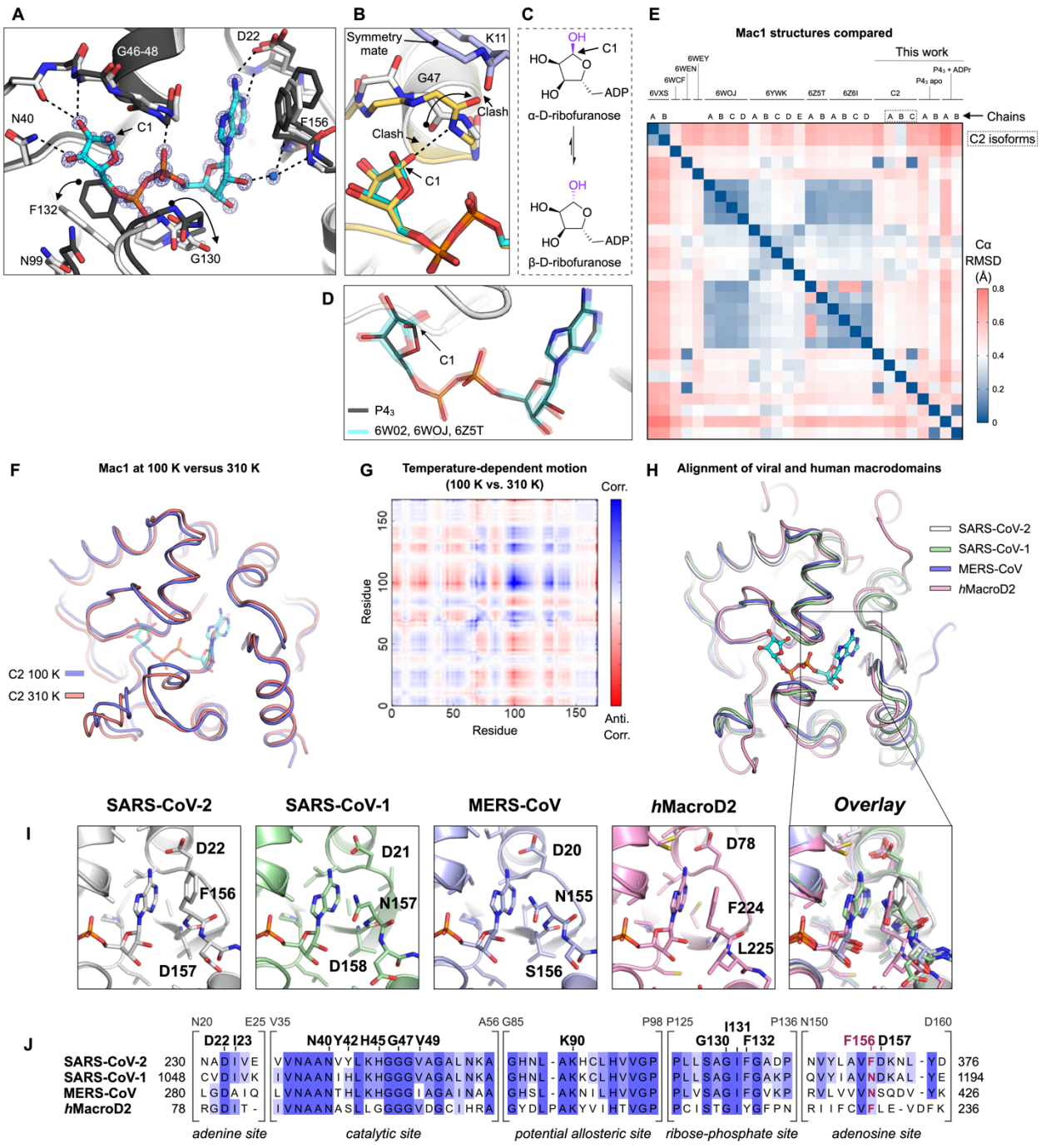
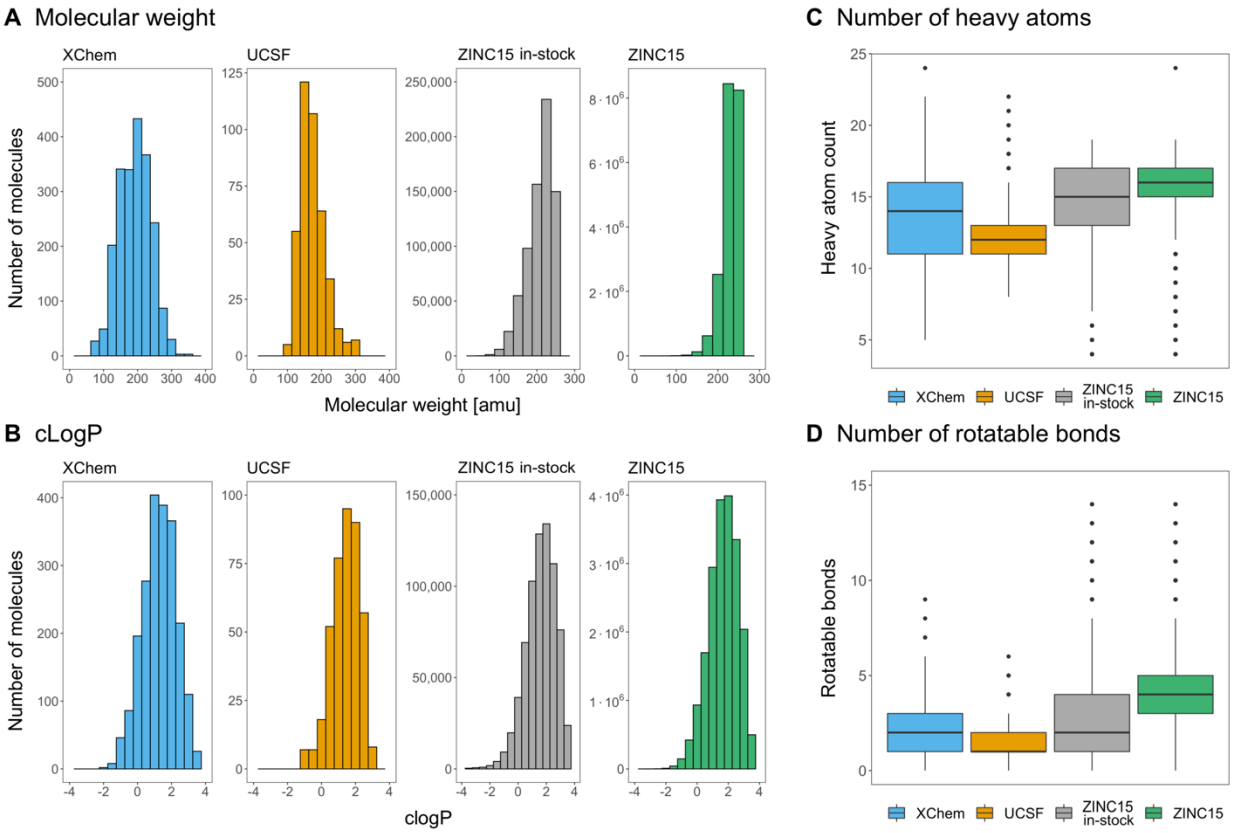


Figure S4. Structure and sequence comparison of Mac1 with related viral and human macrodomains.

A) The structural changes previously reported to occur upon ADP-ribose binding are captured by the Mac1-ADPr structure determined using $P4_3$ crystals. The apo $P4_3$ structure is shown with dark gray sticks, with arrows indicating the changes in protein conformation upon ADP-ribose binding (white sticks). Electron density ($2mF_O$-DF_C) is contoured at 4 σ (blue mesh). **B)** The α-anomer of the terminal ribose of ADP-ribose was observed in the $P4_3$ crystal form (cyan and white sticks). In previously reported structures (e.g. PDB 6W02, yellow sticks), a flip in Gly47

allows the β-anomer to bind by removing a steric block (red dashed line) and forming a new hydrogen bond (black dashed line). However, the Gly47 flip is incompatible with the $P4_3$ crystal form because it would clash with the Lys11 carbonyl of a symmetry mate (blue sticks). In the α-anomer, the anomeric hydroxyl is orientated away from Gly47, and binding can proceed without the peptide flip. **C)** Interconversion between ribose anomers in solution. **D)** Stick representation showing the previously reported Mac1-ADPr structures (cyan sticks) and the new structure determined using $P4_3$ crystals (grey sticks). The agreement between ADP-ribose is excellent, despite different anomers of the terminal ribose being present (α in the $P4_3$ structure, β in the previously reported structures). **E)** Heatmap showing the Cα RMSD values after Cα alignment for 10 previously reported SARS-CoV-2 Mac1 structures (6VXS, 6W02, 6W6Y, 6WCF, 6WEN, 6WOJ, 6WEY, 6YWK, 6Z5T, 6Z6I) and the new structures reported in this work. **F)** Comparison of SARS-CoV-2 Nsp3 Mac1 structures at 100 K (blue) and 310 K (red). ADP-ribose (cyan sticks) is modeled according to its position in PDB 6W02. **G)** Correlation plot showing structural differences between the 100 K and 310 K structures. To generate the plot, the 100 K and 310 K structures were aligned and difference vectors were calculated between identical Cα atoms in the two structures. The plot shows all pairwise dot products between these difference vectors, revealing the extent to which temperature-dependent changes are correlated across the structure. Positive dot products (positive correlations) are colored blue, while negative dot products (negative correlations) are shown in red. The pattern of positive and negative correlations is characteristic of a hinge-bending motion. **H)** Alignment of three coronavirus macrodomain structures with a human macrodomain (hMacroD2). ADP-ribose from the SARS-CoV-2 structure is shown with cyan sticks. **I)** Comparison of the adenosine binding site highlighting key residues involved in adenine and fragment interaction. The adenine coordination by Phe156 is unique to SARS-CoV-2 amongst betacoronaviruses and replaced in SARS-CoV-1 (PDB 2FAV) and MERS-CoV (PDB 5HOL) with asparagine. Human macrodomains, including MacroD2 (PDB 4IQY), have a phenylalanine in this position and interact with adenine in a similar manner compared to SARS-CoV-2 Nsp3 Mac1. This needs to be considered for achieving inhibitor selectivity for viral over human macrodomains. **J)** Sequence alignment showing conservation of residues in the ADP-ribose, catalytic and potential allosteric sites which are targeted by the fragments. Residue numbers on top refer to the construct residue numbering of SARS-CoV-2 Nsp3 Mac1. Numbers on either end of the alignment are residue numbers in the full-length proteins. The adenine-coordinating Phe156 is highlighted in red.



E Scaffold and chemotype analysis

Fragment Library	Number of molecules	Distinct Bemis-Murcko scaffolds	Number of pyrimidines	Number of anions
XChem	2,126	809	72 (3.39%)	115 (5.4%)
UCSF	411	179	12 (2.92%)	144 (35%)
ZINC15 (in-stock)	722,963	69,244	41,531 (5.74%)	85,398 (11.8%)
ZINC15	20,006,175	803,333	890,199 (4.44%)	739,184 (3.7%)

Figure S5. Physical properties, scaffold and chemotype analysis of screened fragment libraries.

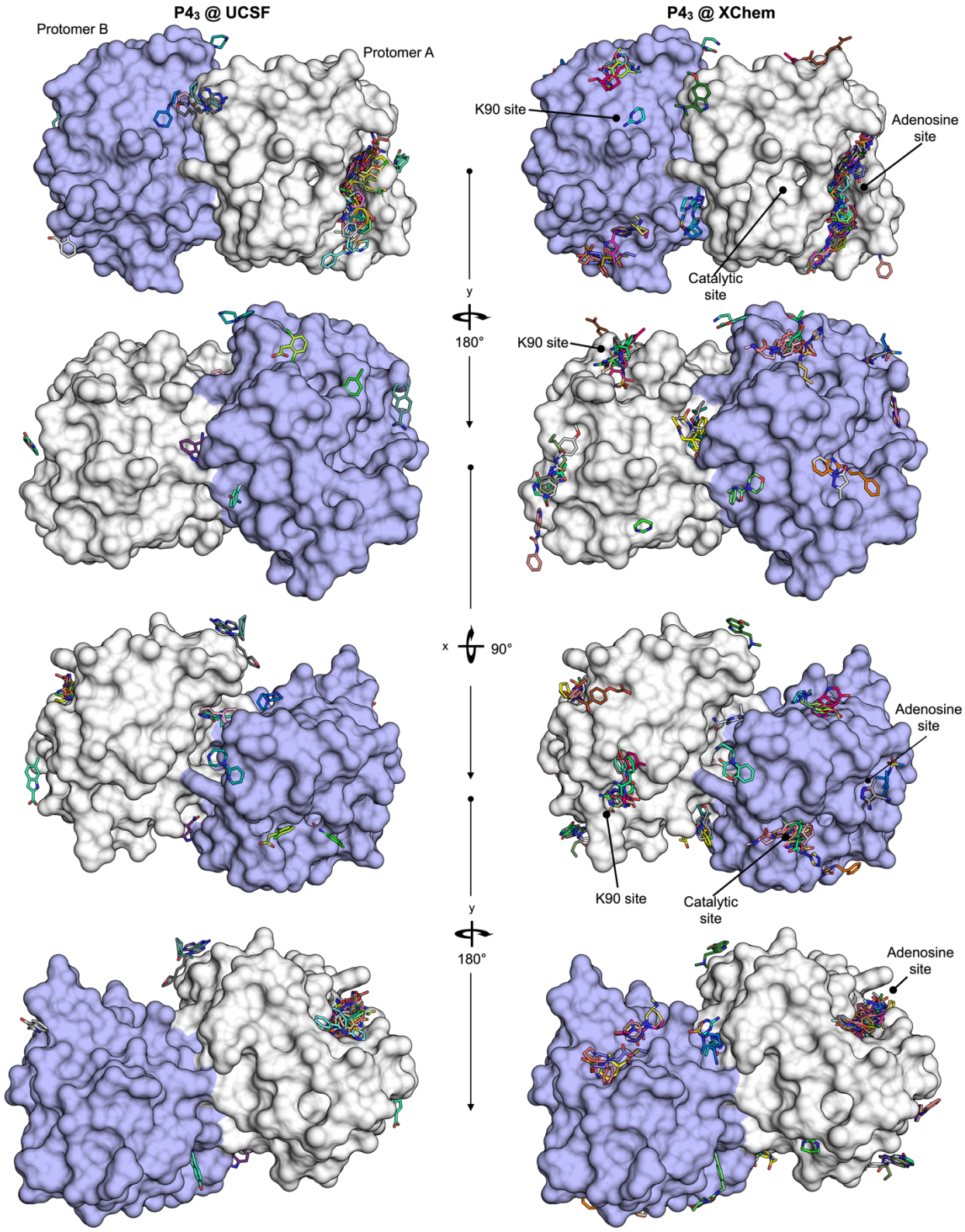


Figure S6. Overview of fragment binding to protomer A (white surface) and protomer B (blue surface) of the $P4_3$ crystals.

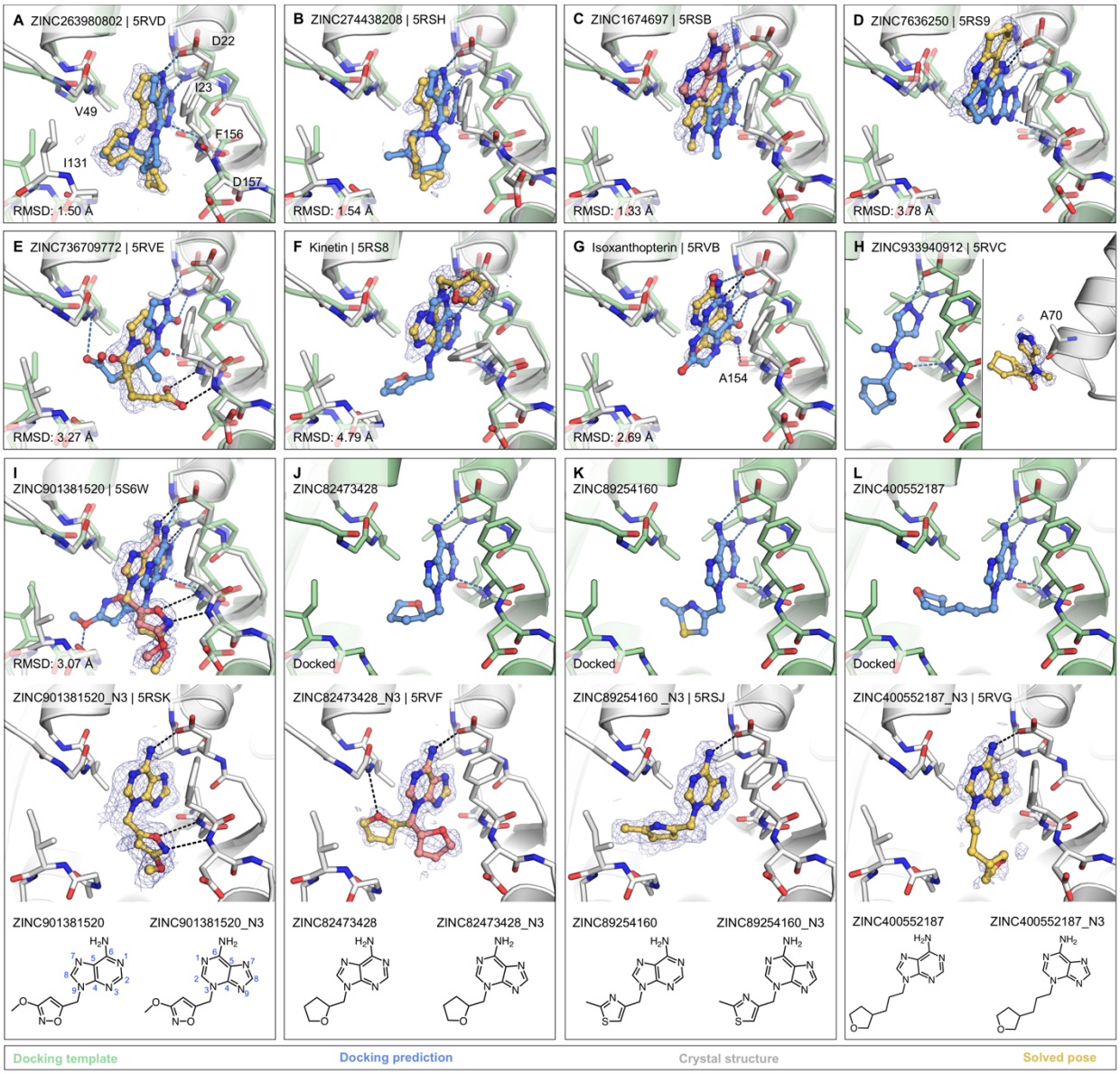


Figure S7. Additional soaking hits from docking and adenine-N3 vs -N9-alkylated isomers. The protein structure (PDB 6W02) (*34*) prepared for virtual screens is shown in green, predicted binding poses are shown in blue, the crystal protein structures are shown in grey, and the solved fragment poses are shown in yellow, with alternative conformations shown in light pink. PanDDA event maps are shown as a blue mesh. Protein-ligand hydrogen bonds predicted by docking or observed in crystal structures are colored light blue or black, respectively. Hungarian RMSD values are presented between docked and crystallographically determined ligand poses.

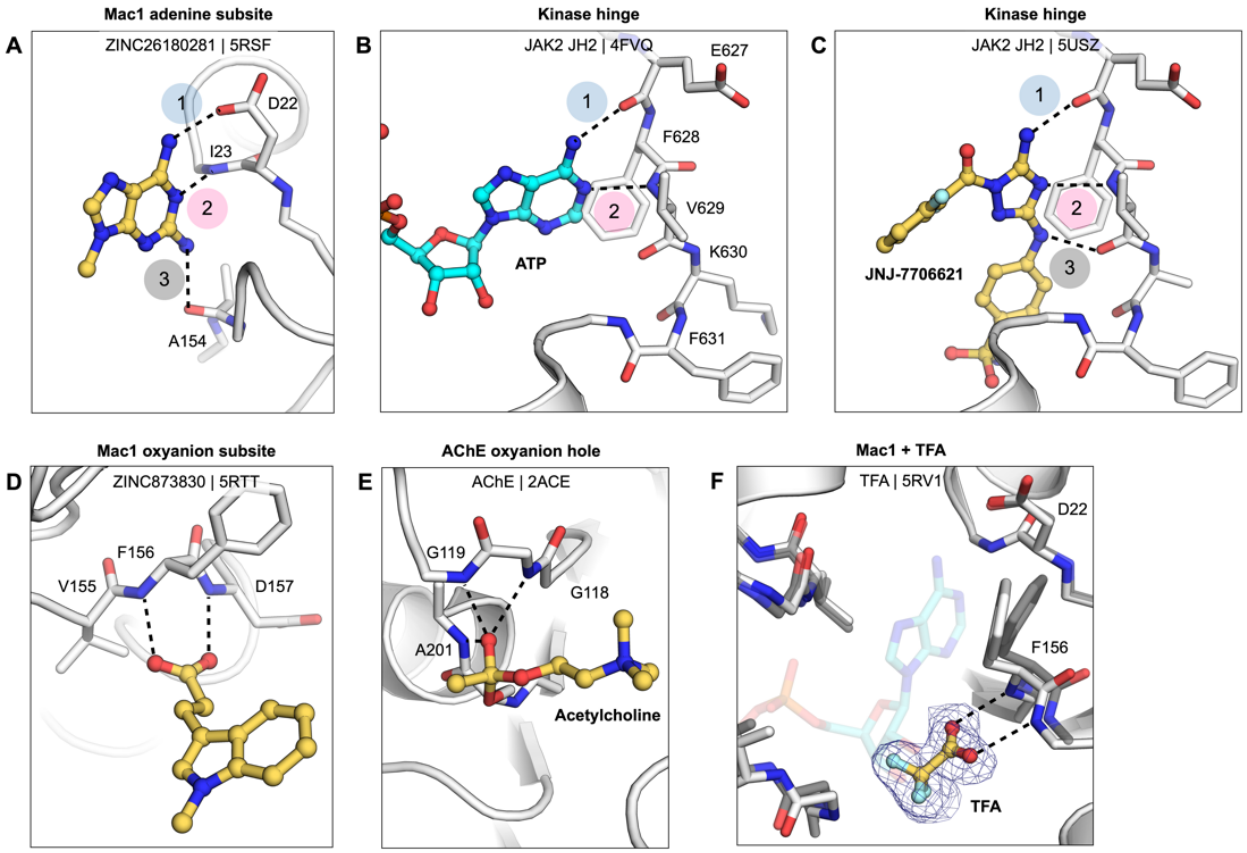


Figure S8. Mac1 subsites compared to the adenine binding subsite in kinases and the oxyanion binding site in carboxylesterases.

A) Key features of the Mac1 adenine subsite are illustrated by the structure of ZINC26180281 (PDB 5RSF). Hydrogen bonds are formed between the C6-amine of the adenine scaffold and the backbone nitrogen of Ile23, and between N1 of the adenine scaffold and the side-chain carboxylate of Asp22. The C2 amine of ZINC26180281 forms a non-canonical hydrogen bond to the backbone carbonyl oxygen of Ala154. **B)** Adenine recognition is similar in the pseudokinase domain of JAK2, however, the C6-amine forms a hydrogen bond to a backbone carbonyl oxygen rather than a side-chain carboxylate. Adenine binding occurs at the hinge residues that connect the N- and C-terminal lobes of the catalytic domain. Interactions that mimic adenine binding to the hinge residues are conserved in the majority of kinase inhibitors (*43*). Like ZINC2618028, kinase inhibitors exploit non-canonical hydrogen bonds. The 1,2,4-triazole derived inhibitor shown in **(C)** forms a hydrogen bond to the backbone carbonyl oxygen of Lys630. **D)** The fragment screens against Mac1 identified 47 oxyanions binding to the backbone nitrogens of Phe156 and Asp157. A comparable oxyanion recognition motif is present in acetylcholinesterase (AChE) **(E).** In AChE, this motif stabilizes negative charge on the oxyanion transition state. **F)** Trifluoroacetic acid (TFA), present as a counter ion for ZINC3860798, was clearly defined in PanDDA event maps binding to the oxyanion subsite. TFA was also observed binding to the oxyanion subsite for fragments ZINC35185198 and ZINC51658946. The docking fragment ZINC263392672 also contained TFA, but no TFA was observed in the oxyanion subsite (PDB 5RSG).

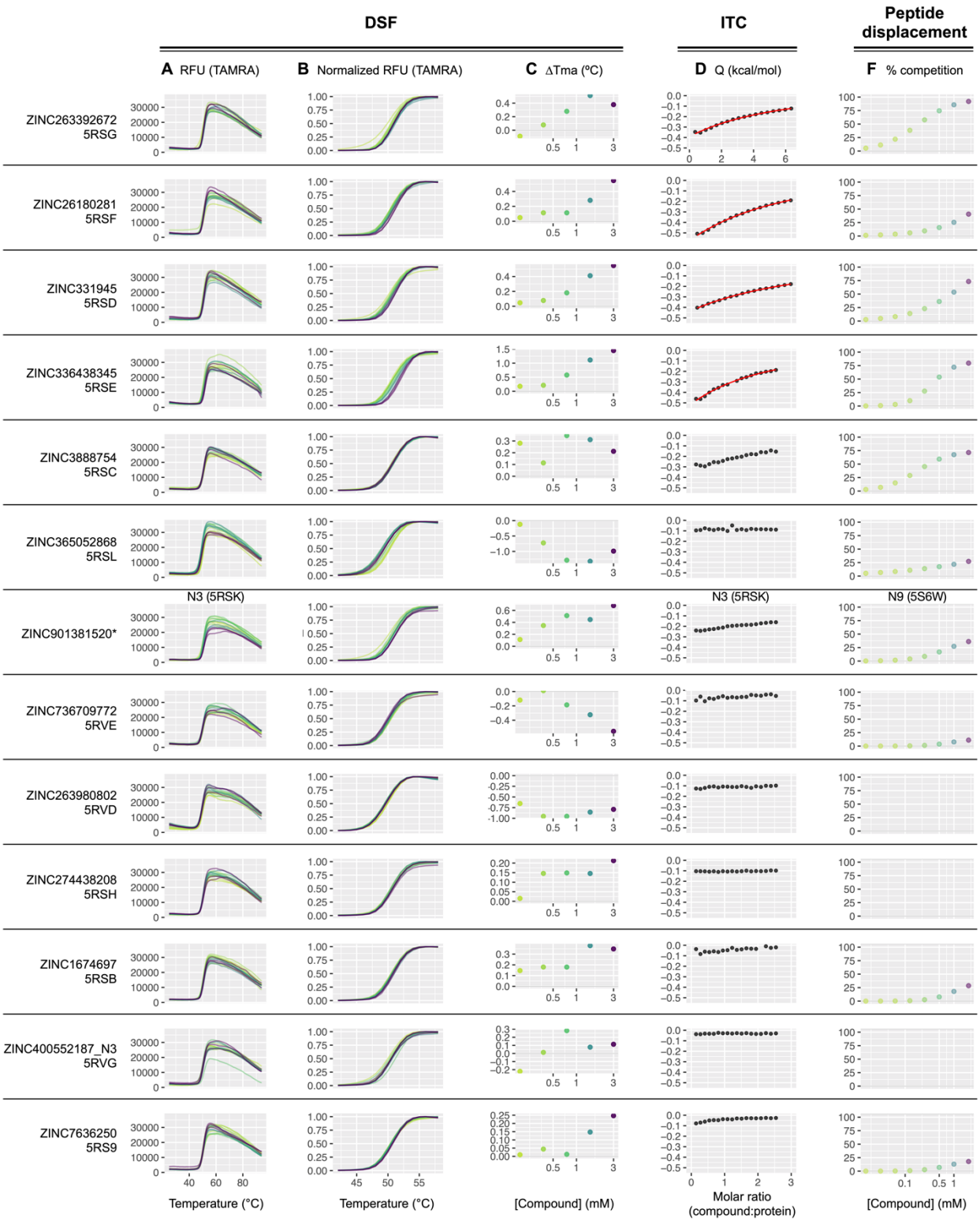


Figure S9. Comparison of DSF, HTRF, and ITC results for compounds tested in all assays. **A)** Raw, un-normalized DSF data for the full measured temperature range (25-94 ºC) demonstrates the absence of confounding changes in curve shape for all compounds. **B)** Normalized raw DSF data, enlarged to visualize compound-induced thermal shifts. **C)** Changes

in Tm_a observed in the presence of 0-3 mM compound. **D)** Integrated heat peaks as a function of compound:protein molar ratio. The red line represents a non-linear fit using a single-site binding model. **E)** Dose-response curves showing competition of the fragments with an ADPr-conjugated peptide for Mac1 binding. (*) ZINC901381520_N3 was tested in DSF and ITC, ZINC901381520_N9 was tested in the peptide displacement assay.

Data file S1. (separate file)
Excel spreadsheet with 1) summary of fragment screens, 2) amino acid sequences of constructs used for crystallography, 3) summary of the data collection strategy for the X-ray diffraction experiments, 4) data reduction and refinement statistics for all X-ray crystal structures reported in this work, 5) summary of all X-ray diffraction datasets collected, 6) classification of all fragment hits, and 7) solution binding data for selected fragments.

Data file S2. (separate file)
PDF with 1) summary of all fragments binding in the adenosine, catalytic and K90 sites of Mac1, 2) DSF data for all compounds tested, 3) ITC data for all compounds tested, and 4) HTRF peptide-displacement data for all compounds tested.

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